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RESEARCH ARTICLE

Feasibility and outcomes of crossflow membrane plasmapheresis for long COVID in Suriname: a randomized double-blind placebo-controlled pilot trial

[version 1; peer review: awaiting peer review]

Rosita Bihariesingh-Sanchit[1,2], Rakesh Bansie[3], Prija Paltoe[4], Rocade Ma[1], Debra Bustamente[1], Dhiradj Ramesar[3], Radha Raghosing-Sanchit[5], John Codrington[6], Inhya Ma-Bihariesingh[2], Stephen Vreden[7], Fey van der Dijs[8], Rishi Mangroo[3], Monique Simson[9], Navin Ramdhani[2], Cesar Fung A Foek[1], Angélique van 't Wout[10,11], Dimitri Diavatopoulos[12,13], Marien de Jonge[12,13], Arno Nierich[1,2,14]

[1]Anesthesiology, Academic Hospital Paramaribo, Paramaribo, Paramaribo District, Suriname
[2]Intensive Care, Academic Hospital Paramaribo, Paramaribo, Paramaribo District, Suriname
[3]Internal Medicine, Academic Hospital Paramaribo, Paramaribo, Paramaribo District, Suriname
[4]Pediatrics, Academic Hospital Paramaribo, Paramaribo, Paramaribo District, Suriname
[5]Dutch Quarter Clinic, Dutch Quarter, Sint Maarten, Netherlands Antilles
[6]Laboratory Science, Academic Hospital Paramaribo, Paramaribo, Paramaribo District, Suriname
[7]Foundation for the Advancement of Scientific Research in Suriname, Paramaribo, Suriname
[8]Health Care Laboratory, Cole Bay, Sint Maarten, Netherlands Antilles
[9]Pulmonology, Academic Hospital Paramaribo, Paramaribo, Paramaribo District, Suriname
[10]AlphaBiomics Limited, Newcastle Upon Tyne, UK
[11]van 't Wout Pharma Consulting, Amsterdam, The Netherlands
[12]Laboratory Medicine - Medical Immunology, Radboud University Medical Center, Nijmegen, The Netherlands
[13]Radboud Community for Infectious Diseases, Radboud University Medical Center, Nijmegen, The Netherlands
[14]HemoClear BV, Zwolle, The Netherlands

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Abstract

Background

Long COVID is heterogeneous and currently without cure. The disease resembles myalgic encephalomyelitis/chronic fatigue syndrome, which is driven by autoantibodies, leading to exploration of therapeutic plasma exchange (TPE) to reduce autoantibody levels as a potential treatment option.

Methods

Open Peer Review

Approval Status *AWAITING PEER REVIEW*

Any reports and responses or comments on the article can be found at the end of the article.

We conducted a randomized, double-blind, placebo-controlled clinical trial comparing TPE with placebo treatment in a resource-limited setting in Suriname. Participants were adults aged 18 to 65 years with physician-confirmed long COVID and questionnaire-defined functional impairment. Treatment consisted of 5 TPE or placebo treatment (blood draw followed by whole blood reinfusion) sessions each separated by two days. Clinical outcomes were assessed using three different fatigue scores at baseline, day 28 and day 90.

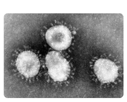
Results

Of 189 patients assessed for eligibility, 18 were randomized and analyzed (mean age [SD], 46 years; 11 females [61%]; mean baseline Chalder fatigue score, 21). While Chalder fatigue scores at 90 days were significantly improved over baseline in the TPE group (mean [SD] at baseline: 20, 90 days: 10, $P = 0.0003$) and not in the placebo group (baseline: 22, 90 days: 12, $P = 0.0628$), between both groups the scores were not statistically significantly different at 90 days ($P = 0.8770$). There was also no significant difference in the percentage of patients that achieved >30% Chalder fatigue score improvement (9 of 11 TPE-treated patients (81.8%) versus 5 of 7 placebo-treated patients (71.4%); difference 10.4% [95% CI, -33.4% to 51.8%]; $P > .9999$). Results were consistent among the three different fatigue scores used. No serious treatment-emergent adverse events were reported in either group.

Conclusions

Our study found reductions in fatigue in both the TPE- and placebo-treatment groups. Larger randomized trials with biomarker-guided patient selection may be required to demonstrate a beneficial effect of TPE in long COVID.

Keywords
Long COVID, therapeutic plasma exchange, immunoadsorption, autoantibodies, randomized clinical trial, resource-limited setting



This article is included in the Coronavirus (COVID-19) collection.

Corresponding author: Rosita Bihariesingh-Sanchit (bihariesinghr@gmail.com)

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Competing interests: Arno P. Nierich is the inventor of the HemoClear filter, holds stock ownership in HemoClear BV, Ceintuurbaan 28, 8024 AA Zwolle, Netherlands and is not involved in patient treatment in Suriname. All other authors: no conflict of interest.

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Introduction

Many patients with COVID-19 infection continue to have non-specific symptoms months after infection, such as persistent fatigue, cognitive issues, dyspnea, headaches, myalgias, sleep disturbances, anosmia, ageusia, and post-exertion malaise.[1] These ongoing symptoms are referred to as long COVID. The rising number of long COVID cases in the absence of validated effective treatments poses a challenge to global health systems.

There are likely multiple, potentially overlapping, causes of long COVID that may include inadequate immune responses, autoimmunity, persistence of pro-inflammatory biomarkers, endothelial and mitochondrial dysfunction, and changes in the gut microbiota.[2,3] Long COVID shares clinical features with other post-viral conditions such as myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS).[4] In both conditions, there is emerging evidence that agonistic receptor autoantibodies directed against neurotransmitter receptors, particularly beta-adrenergic (β1 and β2) and muscarinic (M3 and M4), may contribute to disease pathophysiology, by dysregulating autonomic and vascular signaling pathways.[5,6] Recent IgG transfer experiments have shown that tissue-targeting autoantibodies obtained from a subset of long COVID patients can directly drive symptoms such as pain, fatigue, or neurocognitive problems in mice.[7,8]

Therapeutic plasma exchange (TPE), in which the patient's plasma is removed and replaced with substitution fluid, can reduce the levels of unwanted molecules, such as autoantibodies.[9,10] Although it is an invasive procedure, TPE rarely causes adverse effects and, when they occur, they are usually mild and reversible.[11] Small pilot studies in ME/CFS patients testing positive for autoantibodies suggest that extracorporeal therapies such as TPE may indeed improve clinical symptoms. However, this evidence is limited to small, uncontrolled studies and case-reports.[12–14]

Similarly, case reports[15,16] and three larger non-placebo-controlled pre-post studies of TPE[17,18] and IgG immunoadsorption[19] in long or post COVID patients have reported reductions in autoantibody levels and concurrent improvements of the various measures of long COVID symptoms in most patients. However, these studies were small and/or non-placebo controlled. In contrast, a recent randomized, double-blind, placebo-controlled trial of TPE in 50 long COVID patients found no differences between both groups.[20] We here report the results of our randomized, double-blind, placebo-controlled study of TPE using a novel gravity-driven microfiltration device in 18 long COVID patients in a resource-limited setting in Suriname.

Materials & Methods

This work is reported in adherence to the CONSORT reporting guidelines. This randomized, double-blind, placebo-controlled trial was performed at the Academic Hospital Paramaribo, Paramaribo, Suriname. Eighteen adult patients were included.

Trial design and oversight

In this prospective study, we compared TPE with placebo treatment consisting of conventional blood draw followed by reinfusion of whole blood. The study flow chart in Figure 1 illustrates the study enrolment and design. Patients having experienced RT-PCR confirmed SARS-CoV-2 infection between 2020 and 2022 were identified from hospital records, through social media and radio announcements, or pulmonologist referrals and were contacted to assess eligibility and willingness to participate in the study. Patients were considered for inclusion if they had physician-confirmed long COVID with no symptoms before SARS-CoV-2 infection, questionnaire-defined functional impairment (Chalder fatigue score >10, SF-36 score <65) and >40% working hour reduction but were not bed-ridden. Other inclusion criteria included written informed consent and age between 18 and 65. Exclusion criteria were unwillingness or inability to provide informed consent, history of asthma or any other pathology prior to COVID that could be confused with long COVID, current treatment with anticoagulation or immunosuppression, contraindications to plasmapheresis (lack of peripheral venous access, unstable cardiac pathology), pregnancy or breastfeeding, inability to travel.

Patient and public involvement

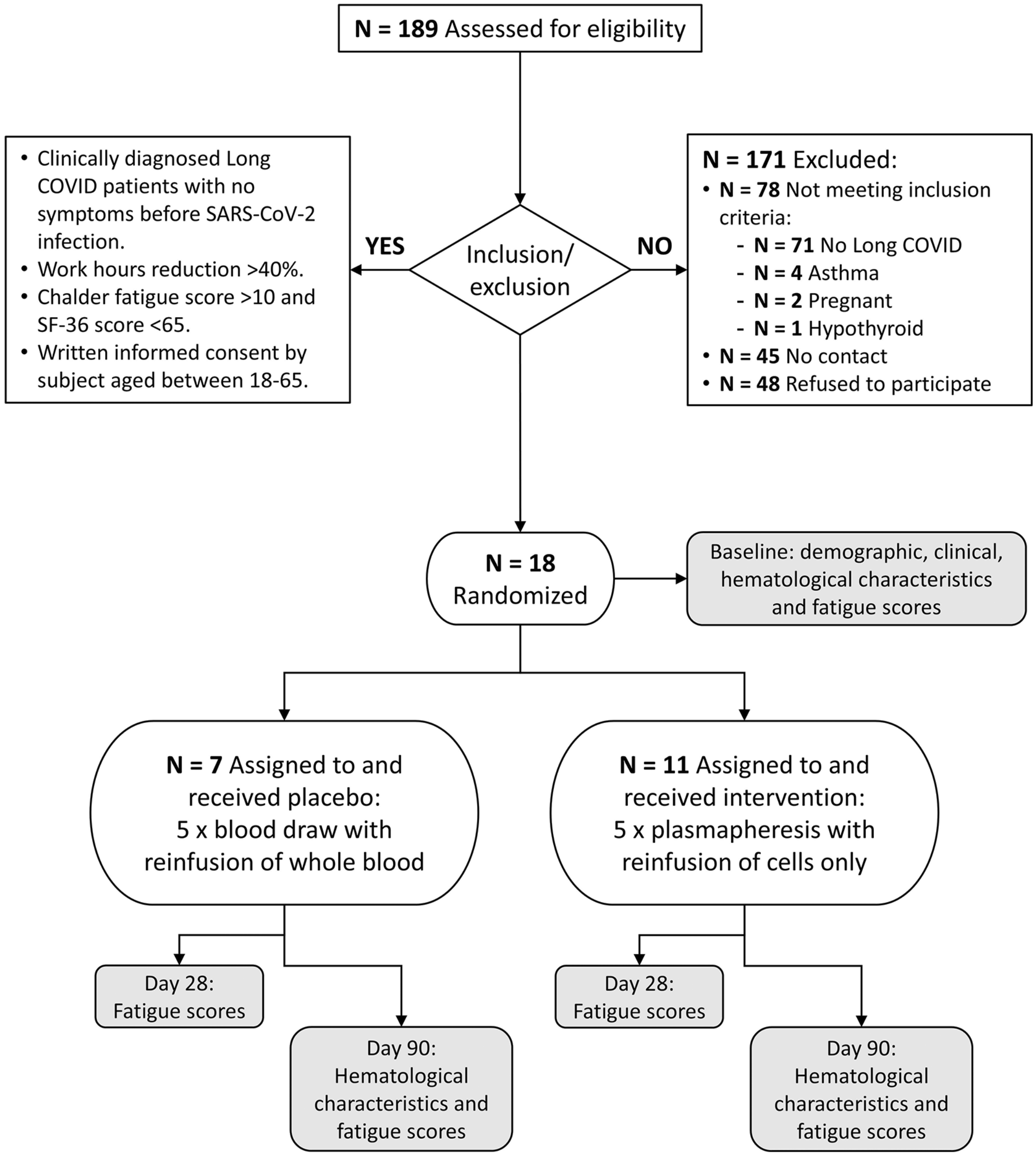
Patients or members of the public were not involved in the design, or conduct, or reporting, or dissemination plans of the research.

Research ethics approval

Ethical approval was granted by the Suriname Ministry of Health's Ethics Review Board (registration no. CMWO:09/2024, 6 April 2024) and registered at the ISRCTN registry (registration no. 13629437, https://doi.org/10.1186/ISRCTN13629437, 10 June 2024).

Informed consent

All patients provided informed written informed consent for participation in the study and the use of their data.



Measurements
Demographics: age, gender, smoking, occupation
Clinical: admission, medication, SARS-COV period
Fatigue scores: SF-36, Chalder, FAS
Hematology - cells: leucocytes, thrombocytes, hemoglobin and hematocrit
Hematology - metabolites/proteins/electrolytes: glucose, urea, creatinine, albumin, sodium, potassium

Figure 1. Patient selection flowchart with study enrolment and experimental design.

Patient population

Participants were recruited at the Academic Hospital Paramaribo in Suriname from April 2024 to January 2025. Of 189 eligible individuals identified, 45 could not be contacted, 48 refused to participate, and 78 did not pass all inclusion and exclusion criteria (71 did not experience long COVID symptoms, 4 had asthma, 2 were pregnant and 1 had hypothyroidism). Those passing all inclusion and exclusion criteria (N=18) were enrolled in the study and randomized to TPE (N=11) or placebo treatment (N=7). Participants were randomized using block randomization with a planned block size of 10. Allocation concealment was ensured using sequentially numbered, opaque, sealed envelopes, which were prepared by an independent researcher. Envelopes were opened sequentially by the study investigator upon patient inclusion. Recruitment was terminated prematurely due to slower than anticipated enrolment in combination with predefined study timelines and resource constraints. Once sufficient data was obtained to assess feasibility of conducting a placebo-controlled extracorporeal intervention trial in a resource-limited setting and to inform the design of future studies, early termination of this pilot trial was deemed appropriate, resulting in an incomplete second block.

Intervention

Treatment in the TPE group consisted of drawing 2 x 500 mL of whole blood, separation into concentrated blood cells and plasma using the HemoClear gravity-driven microfiltration device (HemoClear, Zwolle, The Netherlands),[21] and subsequent reinfusion of the blood cells after washing and dilution with 0.9% NaCl solution. At the discretion of the physician during the first treatment session, two patients (P001 and P013) received an additional infusion with albumin solution (200 g/L) as a safety measure due to transient hypotension during treatment. Treatment in the placebo group consisted of drawing 500 mL of whole blood and subsequent reinfusion of the whole blood. Both groups received five treatment sessions with two days of rest in between. Fatigue score questionnaires were administered at baseline and 28 and 90 days after start of treatment. Patients were blinded for their treatment procedure and throughout follow-up. The investigators performing the treatment procedures (and therefore not blinded) were not involved in administering the follow-up fatigue score questionnaires. The two-week treatment procedures were administered between June 2024 and April 2025, with follow-up through July 2025.

Adverse events

Serious adverse events were defined as hypotension needing vasopressors, vascular access bleeding, or allergic reactions. Procedure related events were defined as no vascular access for blood draw or plasmapheresis.

Outcomes

The primary end point for the study was long COVID symptomatology improvement at 90 days compared to baseline measured using physical and cognitive questionnaires. The following fatigue scores were used: the Chalder fatigue scale,[22] the 36-Item Short Form Survey Instrument (SF-36),[23] and the Fatigue Assessment Scale.[24] Key secondary outcomes included the percentage of patients showing >30% fatigue score improvement on day 90 and fatigue score improvement on day 28.

Sample size and power calculation

In the absence of relevant TPE trials in long COVID at the time of protocol design and submission, this pilot trial design was based on continuous Chalder scores from the PACE trial.[25] Although the PACE protocol predefined a dichotomous ≥30% improvement in fatigue at day 90 as the primary endpoint, continuous fatigue scores were used for the primary endpoint in this study, aligning with the assumptions underlying the sample size calculation. With a baseline Chalder score of 28 ± 7, 24 participants are needed for a two-sided test with 5% significance level and 80% power to compare TPE with placebo.

Statistical analysis

Analyses were performed on the complete data set without imputation. General descriptive statistics were assessed using GraphPad Prism (version 11.0.0). Differences between the TPE and placebo treatment groups were analyzed with Fisher's exact tests for categorical variables, unpaired t tests with Welch's correction for normally distributed continuous variables, Welch's t-tests on log-transformed data for log-normally distributed variables, and Mann-Whitney tests for non-normally distributed continuous variables. Differences in fatigue scores between time points and treatment groups were analyzed using Dunnett's T3 multiple comparisons test in GraphPad Prism (version 11.0.0). Statistical analysis was conducted between April 2025 and March 2026.

Results

Due to the incomplete recruitment (see Methods section for details), the study was not powered for formal hypothesis testing, and results should therefore be considered exploratory. A total of 18 patients were enrolled in the study, 11 of which (61%) were randomized to receive TPE. Demographic and clinical characteristics are described in Table 1. The mean patient age ($\pm$ standard deviation (SD)) was 46 ± 10 years and 11 were female (61%). The intervention (TPE) and placebo treatment groups were not significantly different for most baseline parameters (demographics, hematological parameters and fatigue scores; see Table 1). The TPE and placebo treatment groups did show significant differences in male sex (64% versus 0%, respectively, Fisher's exact test $P = 0.0128$), hemoglobin levels (8.8 ± 1.0 mmol/L versus 7.5

Table 1. Demographic, clinical and hematological factors of the placebo and plasmapheresis treatment groups at baseline.

Characteristic	Placebo		TPE		*P*-value [1]
	(N=7)		(N=11)		
DEMOGRAPHICS					
Age (yr), mean (SD)	48	(4.8)	45	(12.3)	0.4731 [2]
Male sex, n (%)	0	(0)	7	(64)	**0.0128** [3]
Medication, n (%)	3	(43)	6	(55)	>0.9999 [3]
COVID period, n (%)					0.7511 [3]
2020	1	(14)	0	(0)	
2021	4	(57)	8	(73)	
2022	2	(29)	3	(27)	
HEMATOLOGICAL PARAMETERS					
Hemoglobin (mmol/L), mean (SD)	7.5	(0.39)	8.8	(1.0)	**0.0020** [2]
Hematocrit (%), mean (SD)	37	(1.8)	42	(4.1)	**0.0018** [2]
Leukocytes (10e9/L), mean (SD)	8.2	(2.1)	7.7	(2.4)	0.6755 [2]
Thrombocytes (10e9/L), median (IQR)	340	(293-374)	283	(210-362)	0.3749 [4]
Albumin (g/L), mean (SD)	38.9	(4.4)	40.2	(3.6)	0.5449 [2]
Urea (mmol/L), mean (SD)	4.2	(1.5)	4.3	(1.3)	0.8288 [2]
Creatinine (mol/L), mean (SD)	66	(12)	71	(14)	0.4325 [2]
Glucose (mmol/L), median (IQR)	5.5	(5.3-6.7)	5.1	(4.8-6.0)	0.2185 [4]
Sodium (mmol/L), mean (SD)	138	(2.6)	138	(2.5)	0.5289 [2]
Potassium (mmol/L), mean (SD)	3.8	(0.44)	3.8	(0.57)	0.6940 [2]
C-reactive protein (mg/L), mean (SD)	0.75	(0.52)	0.65	(0.40)	0.6684 [2]
Thyroid stimulating hormone (mU/L), mean (SD)	1.9	(1.4)	2.5	(2.7)	0.8813 [5]
FATIGUE SCORES					
SF-36 score, mean (SD)	37	(13)	36	(10)	0.9351 [2]
Chalder score, mean (SD)	22	(6.6)	20	(4.3)	0.6555 [2]
FAS score, mean (SD)	37	(6.6)	34	(6.3)	0.3129 [2]

Abbreviations: IQR, interquartile range; SD, standard deviation; TPE, therapeutic plasma exchange.
[1] *P*-values of <0.05 are bold
[2] Unpaired t-test with Welch's correction
[3] Fisher's exact test
[4] Mann-Whitney test
[5] Lognormal Welch's test

$\pm$ 0.39 mmol/L, unpaired t-test $P = 0.0020$) and hematocrit (42% $\pm$ 1.0% versus 37% $\pm$ 1.8%, unpaired t-test $P = 0.0018$). These lower hemoglobin and hematocrit levels in the placebo treatment group are possibly related to the absence of men in this group.[26] The lower values in the placebo group were measured throughout the study: at day 28 as a trend (hemoglobin: 7.9 $\pm$ 1.1 mmol/L versus 7.2 $\pm$ 0.43 mmol/L, unpaired t-test $P = 0.0786$; hematocrit: 39% $\pm$ 1.1% versus 36% $\pm$ 1.1%, lognormal t-test $P = 0.0547$), and at day 90 again significantly lower in the placebo group (hemoglobin: 8.4 $\pm$ 0.76 mmol/L versus 7.1 $\pm$ 0.66 mmol/L, unpaired t-test $P = 0.0022$; hematocrit: 41% $\pm$ 3.4% versus 35% $\pm$ 2.6%, unpaired t-test $P = 0.0012$) (Figure 2).

Primary outcome: fatigue score improvement at day 90

Fatigue scores were determined using questionnaires at baseline and 28 and 90 days after treatment (Figure 3). All three (Chalder, SF-36 and FAS) fatigue scores were significantly improved in the TPE group at 90 days compared to baseline (Chalder: 10 $\pm$ 5 versus 20 $\pm$ 4, $P = 0.0003$; SF-36: 69 $\pm$ 18 versus 36 $\pm$ 10, $P = 0.0004$; FAS: 24 $\pm$ 4 versus 34 $\pm$ 6, $P = 0.0018$), thereby achieving the primary endpoint of the study. In the placebo group, only the SF-36 score was significantly improved at 90 days compared to baseline (SF36: 66 $\pm$ 13 versus 37 $\pm$ 13, $P = 0.0068$; Chalder:

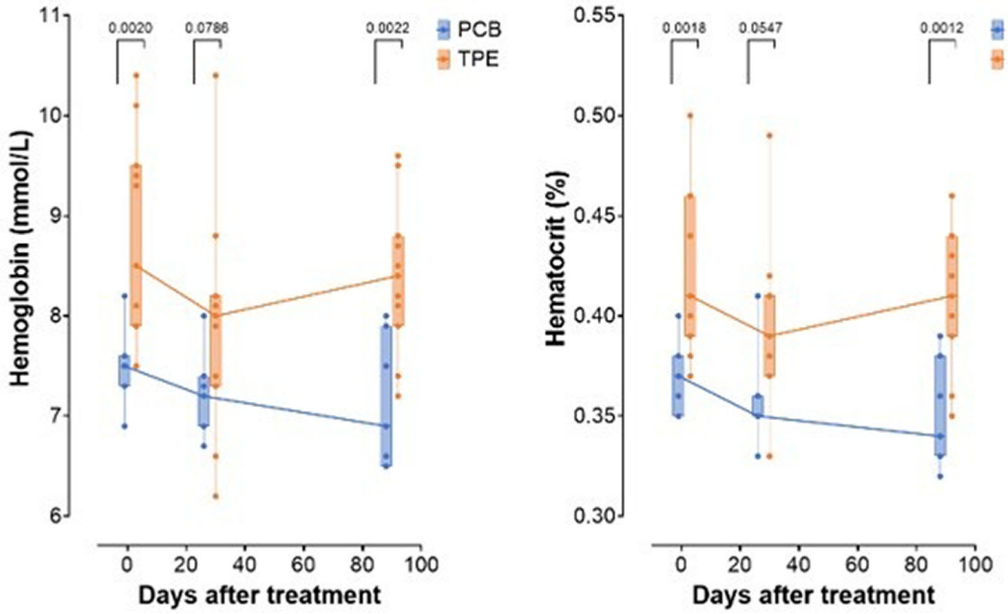


Figure 2. Hemoglobin and hematocrit levels in long COVID patients before and after TPE or placebo treatment. (**A**) Hemoglobin levels before and after treatment with placebo (PCB, n=7, blue) or therapeutic plasma exchange (TPE, n=11, orange). (**B**) Hematocrit levels before and after treatment with PCB or TPE. Boxplot and whiskers depict minimum, 25%, median, 75%, and maximum values with individual points shown as circles. Shown are unpaired t test *P*-values comparing both groups at baseline, day 28 or day 90.

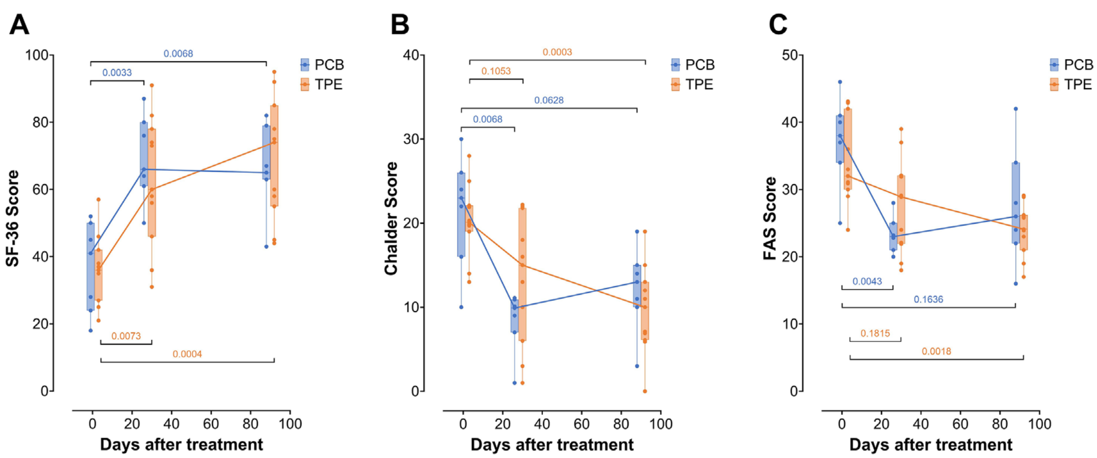


Figure 3. Fatigue scores in long COVID patients before and after TPE or placebo treatment. (**A**) SF-36 scores before and after treatment with placebo (PCB, n=7, blue) or therapeutic plasma exchange (TPE, n=11, orange). (**B**) Chalder scores before and after treatment with PCB or TPE. (**C**) FAS scores before and after treatment with PCB or TPE. Boxplot and whiskers depict minimum, 25%, median, 75%, and maximum values with individual points shown as circles. Shown are Dunnett's T3 multiple comparison test adjusted *P*-values comparing for each group the day 28 and day 90 scores with their respective baseline. At baseline and at day 90, fatigue scores are not significantly different between both groups (*P*-values > 0.86).

12 ± 5 versus 22 ± 7, $P = 0.0628$; FAS: 27 ± 8 versus 37 ± 7, $P = 0.1636$). However, importantly, there were no significant differences in any of the fatigue scores between the TPE and placebo treatment groups at 90 days (Chalder: 10 ± 5 versus 12 ± 5, $P = 0.8770$; SF-36: 69 ± 18 versus 66 ± 13, $P = 0.9988$; FAS: 24 ± 4 versus 27 ± 8, $P = 0.8789$).

Secondary outcome: fatigue score improvement at day 28

In the TPE group, only the SF-36 score was significantly improved at 28 days compared to baseline (SF36: 62 ± 19 versus 36 ± 10, $P = 0.0073$; Chalder: 13 ± 8 versus 20 ± 4, $P = 0.1053$; FAS: 28 ± 7 versus 34 ± 6, $P = 0.1815$) (Figure 3). All three fatigue scores were significantly improved in the placebo group at 28 days compared to baseline (Chalder: 8 ± 4 versus 22 ± 7, $P = 0.0068$; SF-36: 69 ± 13 versus 37 ± 13, $P = 0.0033$; FAS: 23 ± 3 versus 37 ± 7, $P = 0.0043$).

Secondary outcome: percentage with >30% fatigue score improvement

The number of patients achieving at least 30% improvement in their fatigue scores from baseline was plotted over time using Sankey diagrams (Figure 4) and compared between treatment groups at day 28 and day 90 using Fisher's exact tests

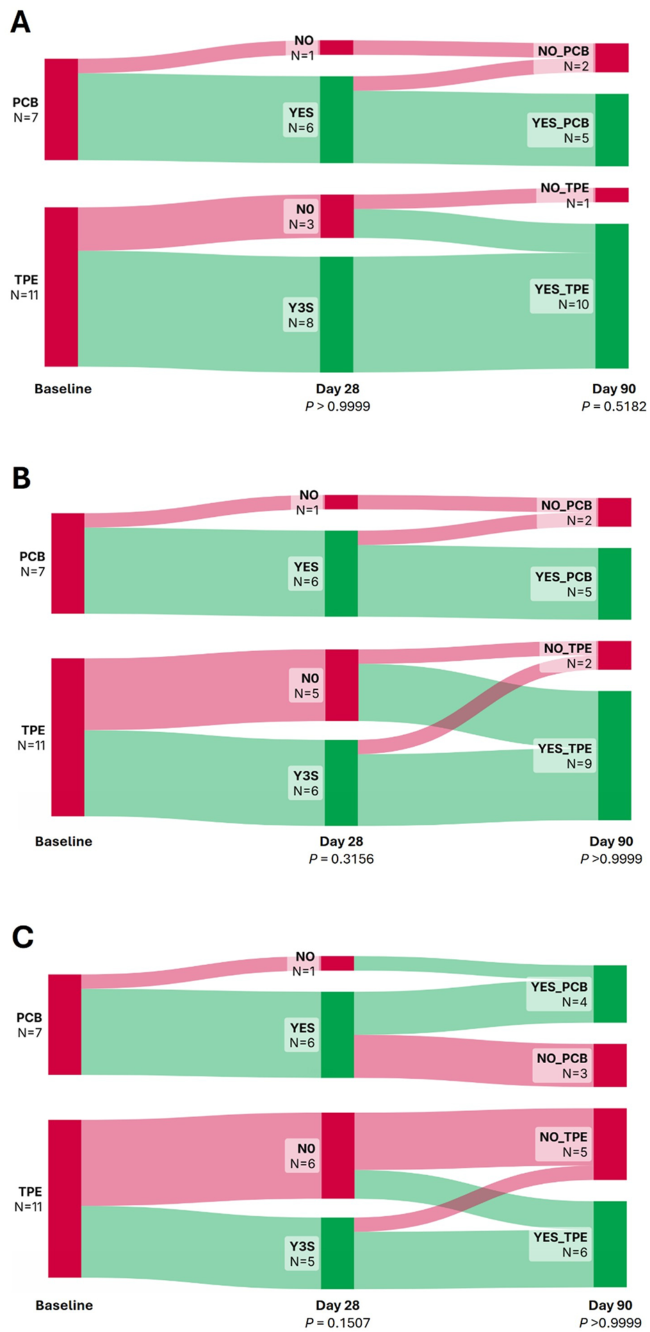


Figure 4. Sankey diagrams showing evolution of fatigue scores over time after TPE or placebo treatment. Shown is >30% score improvement status before and at 28 and 90 days after treatment. Red indicates no score improvement and green indicates >30% score improvement. (**A**) SF-36 score, (**B**) Chalder score, and (**C**) FAS score. Shown are Fisher's exact test *P*-values comparing score improvement at day 28 or day 90 between TPE and placebo groups).

Table 2. Percentage of patients achieving at least 30% fatigue score improvement over baseline at days 28 or 90.

Characteristic	Placebo		TPE		*P*-value [1]
	(N=7)		(N=11)		
SF-36 scores					
Day 28, n (%)	6	(86%)	8	(73%)	>0.9999
Day 90, n (%)	5	(71%)	10	(91%)	0.5282
Chalder scores					
Day 28, n (%)	6	(86%)	6	(55%)	0.3156
Day 90, n (%)	5	(71%)	9	(82%)	>0.9999
FAS scores					
Day 28, n (%)	6	(86%)	5	(45%)	0.1507
Day 90, n (%)	4	(57%)	6	(55%)	>0.9999

Abbreviations: TPE, therapeutic plasma exchange.
[1]Fisher's exact test

(Table 2). No significant differences in fatigue score improvement were found between both treatment groups on either day 28 or day 90.

Secondary outcome: physical and mental fatigue subscores

All three fatigue scores can be divided into a physical and a mental subscore (Figures 5-7). There were no differences in the physical and mental subscore responses compared to the total score responses: the placebo group subscores were both significantly different from baseline at 28 days, the TPE group subscores were both significantly different from baseline

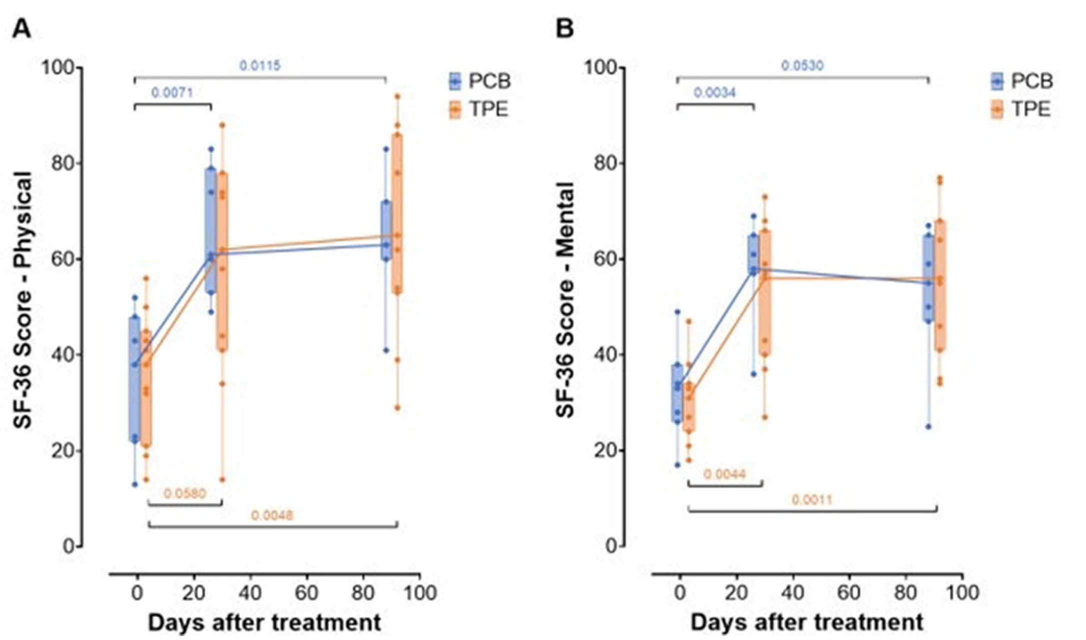


Figure 5. SF-36 physical and mental fatigue scores in long COVID patients before and after TPE or placebo treatment. (**A**) SF-36 physical scores before and after treatment with placebo (PCB, n=7, blue) or therapeutic plasma exchange (TPE, n=11, orange). (**B**) SF-36 mental scores before and after treatment with PCB or TPE. Boxplot and whiskers depict minimum, 25%, median, 75%, and maximum values with individual points shown as circles. Shown are Dunnett's T3 multiple comparison test adjusted *P*-values comparing for each group the day 28 and day 90 scores with their respective baseline. At baseline and at day 90, fatigue scores are not significantly different between both groups (*P*-values > 0.99).

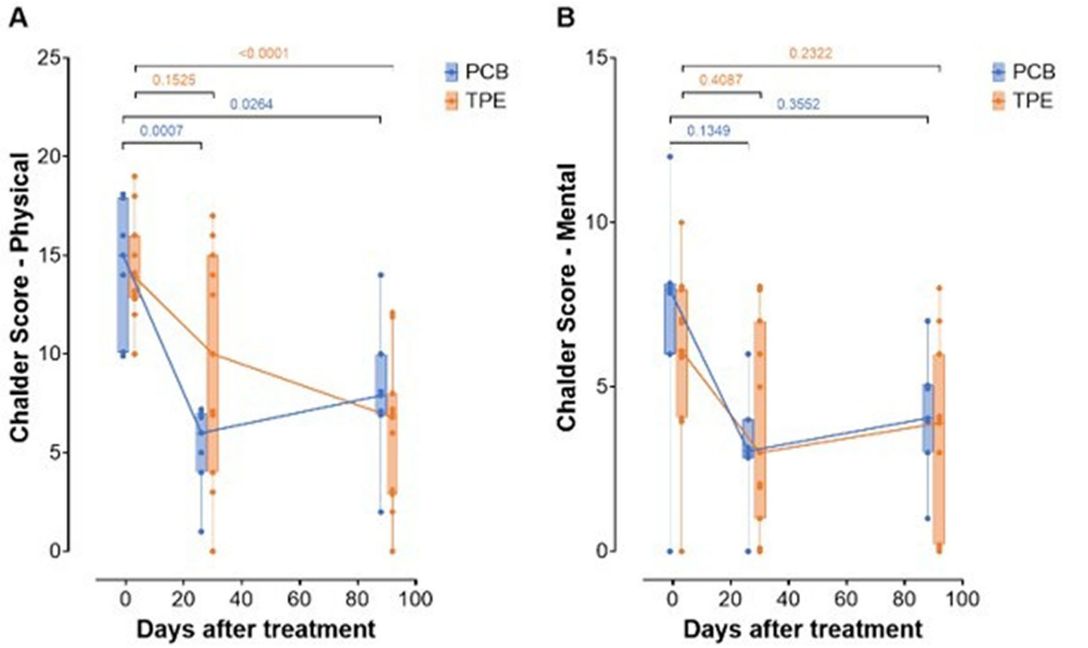


Figure 6. **Chalder physical and mental fatigue scores in long COVID patients before and after TPE or placebo treatment.** (**A**) Chalder physical scores before and after treatment with placebo (PCB, n=7, blue) or therapeutic plasma exchange (TPE, n=11, orange). (**B**) Chalder mental scores before and after treatment with PCB or TPE. Boxplot and whiskers depict minimum, 25%, median, 75%, and maximum values with individual points shown as circles. Shown are Dunnett's T3 multiple comparison test adjusted P-values comparing for each group the day 28 and day 90 scores with their respective baseline. At baseline and at day 90, fatigue scores are not significantly different between both groups (*P*-values > 0.85).

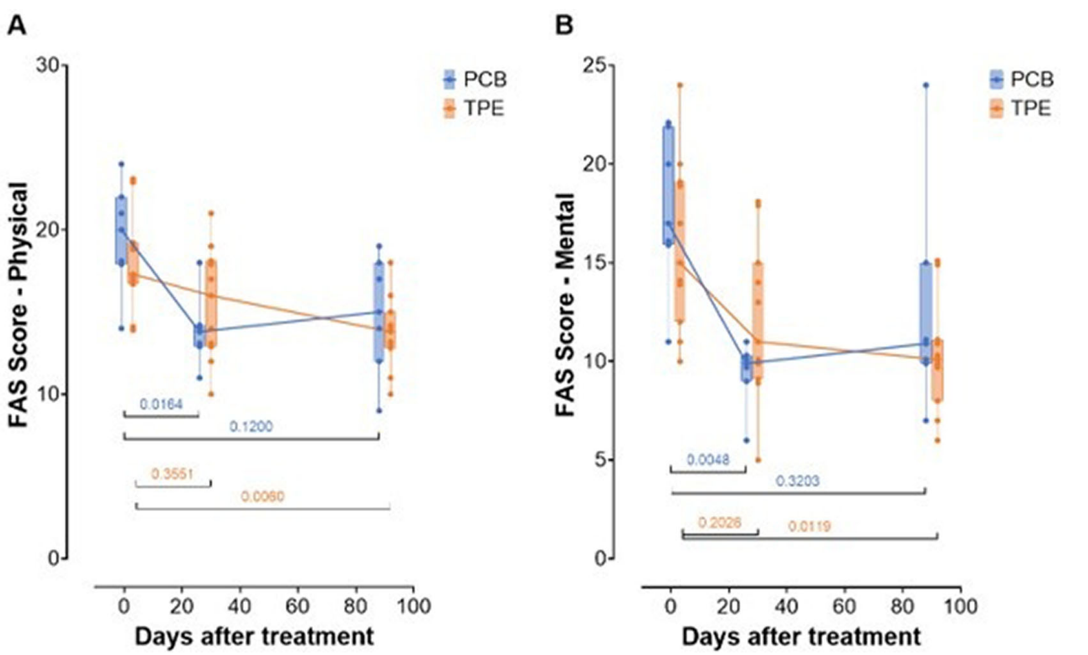


Figure 7. **FAS physical and mental fatigue scores in long COVID patients before and after TPE or placebo treatment.** (**A**) FAS physical scores before and after treatment with placebo (PCB, n=7, blue) or therapeutic plasma exchange (TPE, n=11, orange). (**B**) FAS mental scores before and after treatment with PCB or TPE. Boxplot and whiskers depict minimum, 25%, median, 75%, and maximum values with individual points shown as circles. Shown are Dunnett's T3 multiple comparison test adjusted P-values comparing for each group the day 28 and day 90 scores with their respective baseline. At baseline and at day 90, fatigue scores are not significantly different between both groups (P-values > 0.87).

at 90 days, and there were no significant differences in the subscores scores between the TPE and placebo treatment groups at 90 days.

Secondary outcome: adverse events
No serious adverse events or procedure-related events were recorded. Two patients in the TPE group were hypotensive after the first session which was resolved after infusion with albumin (both patients) and a combination of intravenous ephedrine (5 mg) and phenylephrine (5 mg) and adjustment of the treatment chair (one patient).

Discussion
While long COVID symptoms were significantly improved in our TPE treatment group, symptoms were also significantly improved in the placebo treatment group and there was no difference in mean fatigue score or percentage of patients achieving 30% fatigue score improvement between both groups at follow up. This was true irrespective of the fatigue score used (SF-36, Chalder or FAS) and whether physical or mental fatigue was assessed.

Several case reports and pre-post studies of TPE/immunoadsorption to remove autoantibodies have previously reported reduction in long COVID symptoms concurrent with the reduction in autoantibody levels.[15–19] These studies did not have a placebo treatment arm as a comparator. Indeed, the one study with a placebo treatment arm did not find differences between the TPE treatment and placebo treatment groups.[20]

Importantly, in our study, the placebo treatment group also showed significant fatigue score improvement, especially at the day 28 time point, while the TPE treatment group tended to have the largest fatigue score improvements at the day 90 time point. The magnitude and timing of improvement observed in the placebo group, particularly on day 28, suggest that non-specific treatment effects may play an important role in symptom perception in long COVID. Such effects may include patient expectations, increased clinical attention, and the impact of undergoing an invasive procedure.

Placebo responses are well documented in conditions characterized by subjective symptom burden, including ME/CFS, where improvements in patient-reported outcomes may occur independently of a specific biological intervention.[27,28] Moreover, placebo interventions have been shown to produce clinically meaningful improvements in fatigue across different settings, including cancer-related fatigue and functional disorders, even in the absence of active treatment.[29,30]

Importantly, the invasive nature of the intervention used in this study may have amplified placebo responses. Previous studies have demonstrated that more invasive or complex interventions are associated with stronger placebo effects compared with simple oral treatments likely due to higher expectations of efficacy and the greater perceived intensity of treatment.[28,31] In the present study, both groups underwent repeated blood withdrawal and reinfusion procedures, which may have reinforced these expectations and thereby contributed to the observed clinical improvements, independent of any specific physiological effects of plasma exchange.

Additionally, the placebo intervention in this study, consisting of blood withdrawal and reinfusion, may not have been physiologically inert. Potential effects on circulating factors, hemodynamics, or immune signaling cannot be excluded and may have contributed to symptom improvement. This further complicates the interpretation of treatment effects in studies using procedural control interventions. In addition, the natural course of long COVID may involve gradual symptom improvement over time.[1,32] Longer follow-ups beyond 90 days may resolve whether both groups continue to show significant fatigue score improvement.

The discrepancy between earlier uncontrolled studies reporting clinical improvement following TPE or immunoadsorption[15–19] and the absence of benefit compared to placebo treatment in randomized placebo-controlled trials (this study and[20]) underscores the importance of placebo-controlled study designs in this field. Improvements observed in the pre-post studies may, at least in part, reflect placebo effects, regression to the mean, or natural disease fluctuation rather than true treatment efficacy.

Other differences between the earlier uncontrolled studies and the placebo-controlled trial have been discussed in depth previously[20] and include different plasmapheresis techniques, time elapsed from the acute COVID infection, frequency and volume of plasma exchange and heterogeneity of long COVID sub-phenotypes, including the presence/absence of autoantibodies, included in the study. Here, we used a novel gravity-driven microfiltration device which has been shown to be at least equivalent to conventional centrifugation TPE in terms of quantity and quality of cell salvage in various indications.[21,33,34]

A key limitation of this study is the absence of biomarker measurements to potentially identify more responsive subgroups. Previous uncontrolled studies have suggested that reductions in circulating autoantibody levels or inflammatory markers may correlate with symptom improvement.[17–19] Pathogenic autoantibody effects can vary across biomarker-defined subgroups[7] and may be present in only a subset of patients.[7,8,35] As a result, it remains unclear whether TPE effectively reduced pathogenic factors or whether specific biological subgroups may have derived benefit. Future studies should therefore focus on biomarker-guided patient selection to better identify potential responders.

The study was terminated early due to recruitment constraints, resulting in a limited sample size, which reduces statistical power and increases the risk of type II error. Although no significant clinical differences between groups were observed, small treatment effects cannot be excluded. Larger, adequately powered randomized trials are required to confirm these findings. However, the study does demonstrate the feasibility of conducting randomized controlled trials of extracorporeal therapies in a resource-limited setting.

Conclusions

To conclude, our study found reductions in fatigue in both the placebo and TPE treatment groups. Placebo effects are particularly pronounced in conditions characterized by subjective symptoms such as fatigue, and may be further enhanced by treatment context, patient expectations, and procedural interventions. Larger, adequately powered randomized trials with biomarker-guided patient selection may be required to show beneficial effect of TPE in long COVID. Several of these factors will be addressed in two randomized placebo-controlled trials to evaluate the effect of immunoadsorption on both symptom scores and relevant autoantibodies: IAMPOCO in Germany[36] and TURNLong-COVID in the Netherlands.[37]

Data availability

The materials and data used to support the findings of this study are available at the Zenodo data repository: DOI: 10.5281/zenodo.21466091.[38]

Reporting guidelines

Data repository for: Feasibility and outcomes of crossflow membrane plasmapheresis for long COVID in Suriname: a randomized double-blind placebo-controlled pilot trial. DOI: 10.5281/zenodo.21466091.[38]

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References

1. Davis HE, Assaf GS, McCorkell L, *et al.*: **Characterizing long COVID in an international cohort: 7 months of symptoms and their impact.** *EClinicalMedicine.* 2021 Aug 1; **38**: 101019.
PubMed Abstract | Publisher Full Text | Free Full Text

2. Davis HE, McCorkell L, Vogel JM, *et al.*: **Long COVID: major findings, mechanisms and recommendations.** *Nat. Rev. Microbiol.* 2023 Mar 13; **21**(3): 133–146.
PubMed Abstract | Publisher Full Text | Free Full Text

3. Oba S, Hosoya T, Iwai H, *et al.*: **Long COVID: mechanisms of disease, multisystem sequelae, and prospects for treatment.** *Immunol Med.* 2026 Jan 2; **49**(1): 35–58.
PubMed Abstract | Publisher Full Text

4. Komaroff AL, Lipkin WI: **ME/CFS and Long COVID share similar symptoms and biological abnormalities: road map to the literature.** *Front. Med.* 2023. Frontiers Media SA.
PubMed Abstract | Publisher Full Text

5. Loebel M, Grabowski P, Heidecke H, *et al.*: **Antibodies to β adrenergic and muscarinic cholinergic receptors in patients with Chronic Fatigue Syndrome.** *Brain Behav. Immun.* 2016 Feb 1; **52**: 32–39.
PubMed Abstract | Publisher Full Text

6. Sotzny F, Blanco J, Capelli E, *et al.*: **Myalgic Encephalomyelitis/Chronic Fatigue Syndrome – Evidence for an autoimmune disease.** *Autoimmun. Rev.* 2018; p. 601–609. Elsevier B.V.
PubMed Abstract | Publisher Full Text

7. Chen HJ, Appelman B, Willemen HLDM, *et al.*: **Transfer of IgG from long COVID patients induces symptomology in mice.** *Cell Rep Med.* 2026 Mar; 102693.
Publisher Full Text

8. de Sá KSG, Silva J, Bayarri-Olmos R, *et al.*: **A causal link between autoantibodies and neurological symptoms in long COVID.** *Cell.* 2026 May; **189**(11): 3214–3235.e37.
Publisher Full Text

9. Kambic HE, Nosé Y: **Historical Perspective on Plasmapheresis.** *Ther. Apher.* 1997 Feb 14; **1**(1): 83–108.
Publisher Full Text

10. Rososhansky S, Szymanski IO: **Clinical Applications of Therapeutic Hemapheresis.** *J. Intensive Care Med.* 1996 May 1; **11**(3): 149–161.
Publisher Full Text

11. Altobelli C, Anastasio P, Cerrone A, *et al.*: **Therapeutic Plasmapheresis: A Revision of Literature.** *Kidney Blood Press. Res.*

2023; **48**(1): 66–78.
Publisher Full Text

12. Scheibenbogen C, Loebel M, Freitag H, *et al.*: **Immunoadsorption to remove ß2 adrenergic receptor antibodies in Chronic Fatigue Syndrome CFS/ME. Janigro D, editor.** *PLoS One.* 2018 Mar 15; **13**(3): e0193672.
Publisher Full Text
13. Tölle M, Freitag H, Antelmann M, *et al.*: **Myalgic encephalomyelitis/chronic fatigue syndrome: Efficacy of repeat immunoadsorption.** *J. Clin. Med.* 2020 Aug 1; **9**(8): 1–13.
Publisher Full Text
14. Burgard H: **Clinical Remission After Therapeutic Apheresis in a Patient Suffering from Long Term Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS): A Case Report.** *Int Med Case Rep J.* 2024 Dec; **17**: 997–1002.
Publisher Full Text
15. Kiprov DD, Herskowitz A, Kim D, *et al.*: **Case Report: Therapeutic and immunomodulatory effects of plasmapheresis in long-haul COVID.** *F1000Res.* 2022 Apr 6; **10**: 1189.
Publisher Full Text
16. Giszas B, Reuken PA, Katzer K, *et al.*: **Immunoadsorption to treat patients with severe post-COVID syndrome.** *Ther. Apher. Dial.* 2023 Aug 16; **27**(4): 790–801.
Publisher Full Text
17. Achleitner M, Steenblock C, Dänhardt J, *et al.*: **Clinical improvement of Long-COVID is associated with reduction in autoantibodies, lipids, and inflammation following therapeutic apheresis.** *Mol. Psychiatry.* 2023 Jul 1; **28**(7): 2872–2877.
PubMed Abstract | Publisher Full Text
18. Korth J, Steenblock C, Walther R, *et al.*: **A Single-Center Pilot Study of Therapeutic Apheresis in Patients with Severe Post-COVID Syndrome.** *Horm. Metab. Res.* 2024 Dec 9; **56**(12): 869–874.
PubMed Abstract | Publisher Full Text
19. Stein E, Heindrich C, Wittke K, *et al.*: **Efficacy of repeated immunoadsorption in patients with post-COVID myalgic encephalomyelitis/chronic fatigue syndrome and elevated β2-adrenergic receptor autoantibodies: a prospective cohort study.** *The Lancet Regional Health - Europe.* 2025 Feb 1; **49**.
Publisher Full Text
20. España-Cueto S, Loste C, Lladós G, *et al.*: **Plasma exchange therapy for the post COVID-19 condition: a phase II, double-blind, placebo-controlled, randomized trial.** *Nat. Commun.* 2025 Dec 1; **16**(1).
PubMed Abstract | Publisher Full Text
21. Bihariesingh-Sanchit R, Bansie R, van 't Wout AB, *et al.*: **Therapeutic plasma exchange in critically ill patients in low-income and lower-middle-income countries: medical need and feasibility.** *J. Glob. Health.* 2025; **15**.
PubMed Abstract | Publisher Full Text
22. Cella M, Chalder T: **Measuring fatigue in clinical and community settings.** *J. Psychosom. Res.* 2010 Jul; **69**(1): 17–22.
Publisher Full Text
23. Ware JE, Sherbourne CD: **The MOS 36-item short-form health survey (SF-36). I. Conceptual framework and item selection.** *Med. Care.* 1992 Jun; **30**(6): 473–483.
PubMed Abstract
24. Michielsen HJ, De Vries J, Van Heck GL: **Psychometric qualities of a brief self-rated fatigue measure.** *J. Psychosom. Res.* 2003 Apr; **54**(4): 345–352.
Publisher Full Text
25. White P, Goldsmith K, Johnson A, *et al.*: **Comparison of adaptive pacing therapy, cognitive behaviour therapy, graded exercise therapy, and specialist medical care for chronic fatigue syndrome (PACE): a randomised trial.** *Lancet.* 2011 Mar; **377**(9768): 823–836.
Publisher Full Text
26. **Guideline on haemoglobin cutoffs to define anaemia in individuals and populations.** *Vitamin and Mineral Nutrition Information System.* Geneva: World Health Organization; 2024 [cited 2026 May 4].
Reference Source
27. Cho HJ, Hotopf M, Wessely S: **The Placebo Response in the Treatment of Chronic Fatigue Syndrome: A Systematic Review and Meta-Analysis.** *Psychosom. Med.* 2005 Mar; **67**(2): 301–313.
Publisher Full Text
28. Hróbjartsson A, Gøtzsche PC: **Placebo interventions for all clinical conditions.** *Cochrane Database Syst. Rev.* 2010 Jan 20; **2010**(1).
Publisher Full Text
29. Yennu S, Azhar A, Lu Z, *et al.*: **Open-labeled placebo for the treatment of cancer-related-fatigue in patients with advanced cancer: Results of a randomized controlled trial.** *J. Clin. Oncol.* 2022 Jun 1; **40**(16_suppl): 12006–12006.
Publisher Full Text
30. Fluge Ø, Rekeland IG, Lien K, *et al.*: **B-Lymphocyte Depletion in Patients With Myalgic Encephalomyelitis/Chronic Fatigue Syndrome.** *Ann. Intern. Med.* 2019 May 7; **170**(9): 585–593.
Publisher Full Text
31. Kaptchuk TJ, Kelley JM, Conboy LA, *et al.*: **Components of placebo effect: randomised controlled trial in patients with irritable bowel syndrome.** *BMJ.* 2008 May 3; **336**(7651): 999–1003.
Publisher Full Text
32. Sudre CH, Murray B, Varsavsky T, *et al.*: **Attributes and predictors of long COVID.** *Nat. Med.* 2021 Apr 10; **27**(4): 626–631.
Publisher Full Text
33. Ramatsea NC, Booyens M, Sempa JB, *et al.*: **Feasibility and effectiveness of the novel HemoClear microfiltration cell-salvage system in cesarean sections in South Africa.** *Int. J. Gynecol. Obstet.* 2026 Jan 1; **172**(1): 396–405.
PubMed Abstract | Publisher Full Text
34. Hoetink A, Scherphof SF, Mooi FJ, *et al.*: **An in vitro Pilot Study Comparing the Novel HemoClear Gravity-Driven Microfiltration Cell Salvage System with the Conventional Centrifugal XTRA™ Autotransfusion Device.** *Anesthesiol Res Pract.* 2020; **2020**.
Publisher Full Text
35. Schouten D, Appelman B, den Dunnen J, *et al.*: **Ned T Med Microb - 2025 - De rol van autoantistoffen bij long COVID.** *Nederlands Tijdschrift voor Medische Microbiologie.* 2025 Sep [cited 2026 Apr 3]; **33**(3): 98–102.
Reference Source
36. Stortz M, Klimpke P, Kommer A, *et al.*: **Immunoadsorption study Mainz in adults with post-COVID syndrome (IAMPOCO)—a single-blinded sham-controlled crossover trial to evaluate the effect of immunoadsorption on post-COVID syndrome.** *Trials.* 2025 Apr 3; **26**(1): 119.
Publisher Full Text
37. Wiersinga WJ: *ClinicalTrials.gov.* Amsterdam: 2026 [cited 2026 Apr 3]; p. 1–13. NCT07316127 - Immunoadsorption in autoimmune Long COVID (TURNLongCOVID).
Reference Source
38. Bihariesingh-Sanchit R: **Data repository.** *Zenodo.* 2026.
Publisher Full Text